

# DANIEL J. LLOVERAS

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## INTERESTS

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Atmospheric predictability; deep learning weather prediction; synoptic and mesoscale meteorology; numerical methods

## EDUCATION

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**Ph.D. in Atmospheric Sciences**, University of Washington December 2023  
Certificate in data science

**M.S. in Atmospheric Sciences**, University of Washington March 2021

**B.S. in Marine and Atmospheric Science**, University of Miami, *summa cum laude* May 2018  
Majors in meteorology and applied mathematics, minor in broadcast journalism

## RESEARCH EXPERIENCE

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**Postdoctoral Research Associate**, U.S. Naval Research Laboratory 2024 – present

### *Scientific Experience*

- Investigated intrinsic predictability of mesoscale polar lows
- Explored predictability of synoptic-scale environments associated with polar lows using deep learning sensitivity analysis

### *Technical Experience*

- Ran Coupled Ocean/Atmosphere Mesoscale Prediction System (COAMPS) on high-performance computing (HPC) machines
- Developed Python code for deep learning sensitivity analysis with GraphCast
- Analyzed and visualized model output with Python

**Graduate Research Assistant**, University of Washington 2018 – 2023

### *Scientific Experience*

- Investigated 2–4-day predictability of midlatitude cyclones
- Proposed and tested a hypothesis for why deploying targeted observations to improve midlatitude-cyclone forecasts often does not work

### *Technical Experience*

- Ran COAMPS, COAMPS adjoint, and Weather Research and Forecasting Model (WRF) on HPC machines
- Developed novel Python and Fortran code to generate initial conditions for idealized midlatitude-cyclone simulations
- Analyzed and visualized model output with Python

**Undergraduate Research Assistant**, University of Miami 2016 – 2018

### *Scientific Experience*

- Investigated how shortwave-absorbing smoke particles affect low-cloud properties
- Found that poor data from cloud condensation nuclei (CCN) counter was due to high ambient relative humidity

### *Technical Experience*

- Analyzed data from Layered Atlantic Smoke Interactions with Clouds (LASIC) field campaign using Interactive Data Language (IDL)
- Developed air-drying mechanism to improve CCN data quality

*Scientific Experience*

- Showed that representation of sea-surface temperature over northeast Atlantic Ocean is essential to the seasonal predictability of precipitation over southeastern U.S.

*Technical Experience*

- Analyzed output from Forecast-Oriented Low Ocean Resolution Model (FLOR) with MATLAB
- Computed verification statistics using ERA-Interim reanalysis data
- Conducted composite and empirical orthogonal function (EOF) analysis

**PUBLICATIONS**

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**Lloveras, D. J.** and J. D. Doyle, 2024: Exploring the intrinsic predictability of an Arctic polar low. *Quart. J. Roy. Meteor. Soc.*, submitted.

**Lloveras, D. J.**, J. D. Doyle, and D. R. Durran, 2024: Can observation targeting be a wild goose chase? An adjoint-sensitivity study of a U.S. East Coast cyclone forecast bust. *J. Atmos. Sci.*, **82**, 343–360. <https://doi.org/10.1175/JAS-D-24-0044.1>

**Lloveras, D. J.** and D. R. Durran, 2024: Improving the representation of moisture and convective instability in baroclinic-wave channel simulations. *Mon. Wea. Rev.*, **152**, 1469–1486. <https://doi.org/10.1175/MWR-D-23-0210.1>

**Lloveras, D. J.**, D. R. Durran, and J. D. Doyle, 2023: The two- to four-day predictability of midlatitude cyclones: Don't sweat the small stuff. *J. Atmos. Sci.*, **80**, 2613–2633. <https://doi.org/10.1175/JAS-D-22-0232.1>

**Lloveras, D. J.**, L. H. Tierney, and D. R. Durran, 2022: Mesoscale predictability in moist midlatitude cyclones is not sensitive to the slope of the background kinetic energy spectrum. *J. Atmos. Sci.*, **79**, 119–139. <https://doi.org/10.1175/JAS-D-21-0147.1>

**PRESENTATIONS**

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**Lloveras, D. J.**, J. D. Doyle, and D. R. Durran, 2025: Can observation targeting be a wild goose chase? An adjoint-sensitivity study of a U.S. East Coast cyclone forecast bust. *4th Symposium on Mesoscale Processes at the 105th American Meteorological Society (AMS) Annual Meeting*. Oral presentation.

**Lloveras, D. J.** and D. R. Durran, 2025: Improving the representation of moisture and convective instability in baroclinic-wave channel simulations. *29th Conference on Numerical Weather Prediction at the 105th AMS Annual Meeting*. Poster presentation.

**Lloveras, D. J.** and J. D. Doyle, 2024: Exploring the intrinsic predictability of an Arctic polar low. *4th Symposium on Mesoscale Processes at the 105th AMS Annual Meeting*. Oral presentation.

**Lloveras, D. J.**, D. R. Durran, and J. D. Doyle, 2023: Initial-condition-error growth in idealized midlatitude cyclones. *20th AMS Conference on Mesoscale Processes*. Oral presentation.

**Lloveras, D. J.**, D. R. Durran, and J. D. Doyle, 2023: Upscale versus large-scale error growth in midlatitude cyclones. *3rd Symposium on Mesoscale Processes at the 103rd AMS Annual Meeting*. Oral presentation.

**Lloveras, D. J.**, D. R. Durran, L. H. Tierney, and J. D. Doyle, 2022: The predictability of midlatitude cyclones: Are butterflies important? *National Center for Atmospheric Research–Mesoscale and Microscale Meteorology Laboratory (NCAR–MMM) Happy Hour Seminar*. Invited oral presentation.

**Lloveras, D. J.**, L. H. Tierney, and D. R. Durran, 2022: Mesoscale predictability in moist midlatitude cyclones is not sensitive to the slope of the background kinetic energy spectrum. *19th Conference on Mesoscale Processes at the 102nd AMS Annual Meeting*. Remote oral presentation.

**Lloveras, D. J.**, L. H. Tierney, and D. R. Durran, 2021: Mesoscale predictability in moist midlatitude cyclones is not sensitive to the slope of the background kinetic energy spectrum. *American Geophysical Union (AGU) Fall Meeting 2021*. Remote poster presentation.

**Lloveras, D. J.** and P. Zuidema, 2018: Assessment of low-cloud changes in the presence of shortwave-absorbing smoke. *17th Student Conference at the 98th AMS Annual Meeting*. Poster presentation.

**Lloveras, D. J.** and X. Yang, 2018: Evaluating the predictability of summertime precipitation over the southeastern United States. *17th Student Conference at the 98th AMS Annual Meeting*. Poster presentation.

## HONORS AND AWARDS

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|-------------------------------------------------------------------------------------|-------------|
| Graduate Student Distinguished Service Award, University of Washington              | 2022        |
| Honorable Mention Oral Presentation, 19th AMS Conference on Mesoscale Processes     | 2022        |
| Outstanding Student Presentation Award, AGU Fall Meeting                            | 2021        |
| Achievement Rewards for College Scientists Fellowship                               | 2018 – 2021 |
| Honorable Mention, National Science Foundation Graduate Research Fellowship Program | 2019        |
| Departmental Honors in Atmospheric Science, University of Miami                     | 2018        |
| Outstanding Graduating Senior in Mathematics, University of Miami                   | 2018        |
| Honor Roll and Dean's List, University of Miami                                     | 2014 – 2018 |
| President's Scholarship, University of Miami                                        | 2014 – 2018 |

## TEACHING EXPERIENCE

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**Instructor**, ATM S 490: Current Weather Analysis, University of Washington  
*Winter Quarter 2021, Autumn Quarter 2021, Spring Quarter 2022*

- Led weekly weather discussions
- Presented lectures on the fundamentals and frontiers of weather analysis and forecasting
- Taught sections for both majors and non-majors

**Teaching Assistant**, ATM S 111: Global Warming, University of Washington  
*Winter Quarter 2022, Spring Quarter 2023*

- Presented lectures on the affect of climate change on hurricanes
- Led weekly quiz sections to review material and facilitate discussions
- Developed new homework and exam questions
- Graded assignments and final projects

**Teaching Assistant**, ATM S 103: Hurricanes and Thunderstorms, University of Washington  
*Spring Quarter 2020*

- Presented weekly lectures on a “storm of the week”
- Led exam review sessions
- Developed new homework and exam questions
- Adapted to the first online-learning quarter of the pandemic

## SERVICE AND OUTREACH

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| <b>Member</b> , AMS Committee on Weather Analysis and Forecasting                                                                                                                                                                                                                                   | 2020 – Present |
| <ul style="list-style-type: none"><li>• Chaired outreach subcommittee</li><li>• Planned conferences and chaired sessions</li><li>• Evaluated AMS glossary submissions and award nominees</li><li>• Co-authored 5-year strategic and implementation plans</li></ul>                                  |                |
| <b>Mentor</b> , University of Washington Graduate-Undergraduate Mentoring Program                                                                                                                                                                                                                   | 2019 – 2023    |
| <ul style="list-style-type: none"><li>• Mentored three undergraduates in Department of Atmospheric Sciences</li><li>• Met quarterly to discuss courses and career opportunities</li><li>• Attended quarterly social events to engage with other undergraduates</li></ul>                            |                |
| <b>Volunteer</b> , University of Washington Outreach Program                                                                                                                                                                                                                                        | 2018 – 2023    |
| <ul style="list-style-type: none"><li>• Provided demos during science nights at local elementary schools</li><li>• Hosted field trips from local schools by providing demos and tours of the building</li></ul>                                                                                     |                |
| <b>Manager</b> , University of Washington Weather Challenge Forecasting Team                                                                                                                                                                                                                        | 2020 – 2023    |
| <ul style="list-style-type: none"><li>• Coordinated weekly weather discussions and daily email weather briefings</li><li>• Mentored undergraduates interested in weather forecasting and analysis</li><li>• Recruited and registered team members in the national forecasting competition</li></ul> |                |
| <b>Graduate President</b> , University of Washington Student Chapter of the AMS                                                                                                                                                                                                                     | 2020 – 2022    |
| <ul style="list-style-type: none"><li>• Organized monthly social and professional development events</li><li>• Raised funds for undergraduates to attend AMS Student Conference</li><li>• Mentored undergraduate officers</li></ul>                                                                 |                |
| <b>Treasurer</b> , University of Miami Student Chapter of the AMS                                                                                                                                                                                                                                   | 2017 – 2018    |
| <ul style="list-style-type: none"><li>• Managed the chapter's budget</li><li>• Coordinated reimbursements for travel and events</li></ul>                                                                                                                                                           |                |

Last updated: February 2025